Code No: 113BR **R13**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B.Tech II Year I Semester Examinations, September/October - 2023
BASIC ELECTRICAL ENGINEERING
(Common to CSE, IT)

Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

PART- A

(25 Marks)

1.a) What are the differences between independent and dependent sources? [2]
b) State and explain the maximum power transfer theorem. [3]
c) Define power factor of an a.c circuit. [2]
d) The impedance of an electrical circuit is (30 -j50) ohms. Determine (i) the resistance, (ii) the capacitance, and (iii) the magnitude of the impedance, when the circuit is connected to a 240 V, 50 Hz supply. [3]
e) Define transformation ratio of transformer. [2]
f) What is the working principle of transformer? [3]
g) What is the condition for maximum efficiency of a DC machine? [2]
h) What are the different types of DC Generators? [3]
i) Differentiate gravity and spring control mechanisms used in measuring instruments. [2]
j) What are the advantages and limitations of MI type instruments. [3]

PART- B

(50 Marks)

2.a) Using method of superposition, determine the current through the 5kΩ resistors for the circuit in Figure 1 .

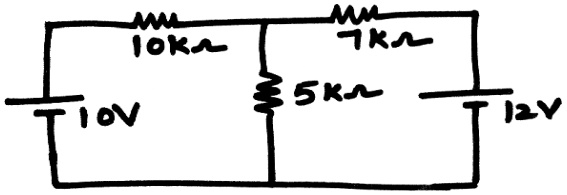


Figure: 1

b) Explain the Kirchhoff's laws. [6+4]

OR

3.a) State and explain Thevenin's theorem.
b) Determine the current I in the network shown in Figure 2 by using Thevenin's theorem. [3+7]

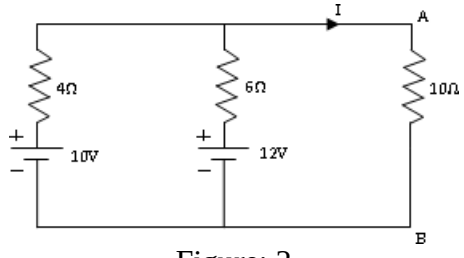


Figure: 2

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4. An alternating voltage expressed as $i = 14.14 \sin(314t+\varphi)$
Determine a) Time period b) Frequency
c) rms value of current d) Average value of current
e) peak factor f) form factor [10]

OR

5. A resistance of 10Ω connected in series with a 50mH inductance across a 230V, 50Hz supply.
Find a) impedance b) current c) power factor
d) Phase angle e) voltage across resistor and inductor [10]

6.a) Describe how the open circuit test is conducted on the single phase transformer.
b) Explain about transformer regulation. [7+3]

OR

7.a) Explain about various losses of Single phase transformer? How to minimize them?
b) State the various parts of a transformer and their function. [6+4]

8. Explain the working principle of D.C. Motor with neat sketches. [10]

OR

9.a) Three phase induction motor is wound for four poles and is supplied from 50 Hz supply. Calculate
i) Synchronous speed
ii) the speed of the rotor when the slip is 3% and
iii) the rotor frequency when the speed of the rotor is 900rpm.
b) Enumerate the different types of losses in a DC motor? [6+4]

10.a) Summarize the various effects utilized in different types of measuring Instruments.
b) Explain the working Principe of repulsion type Moving Iron Instrument. [5+5]

OR

11.a) Explain the working of Moving coil Permanent magnet instruments.
b) What are the different damping mechanisms provided to the measuring instruments? Explain in brief. [6+4]

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